

A Derivation of Darcy's law using Homogenisation

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1 Introduction

These notes are heavily based off the book "Introduction to Perturbation Methods" by Holmes [1].

Consider the flow through a porous solid. The flow domain, V consists of an impermeable solid region, denoted V_s and a fluid region, denoted V_f . It is important to note that our system of interest - this fluid-solid matrix - has two different length scales. The first is a macroscale length scale that is associated with the overall size of the system (e.g. the dimensions of the matrix), which we denote ' L .' The second is associated with the underlying microstructure and can be thought of as a pore length scale (e.g. the diameter of the grains in a granular medium), which we denote ' r_p .' We also denote $\varepsilon = r_p/L \ll 1$ as the ratio between the microscale and the macroscale. The possibility of variation on both length scales could be challenging to resolve numerically. In order to circumvent the geometric complexity of the problem and make some analytical progress, we can use an asymptotic technique called the "The method of Homogenisation" to derive effective laws, that are valid on the macroscale, but nonetheless account for the microstructure variation of the flow domain. In essence, the crux of the method lies in smoothing out any variation on the microscale, by averaging all quantities on the microscale. We will implement a multiple scales analysis before introducing the relevant average. We define the macroscale variable $\mathbf{x}$ and the microscale variable $\mathbf{y} = \mathbf{x}/\varepsilon$. As with any multiple scales problem, we will treat x and $\mathbf{y}$ as independent and impose the secularity condition that the solution is exactly periodic in $\mathbf{y}$. Using the chain rule, we can transform derivatives as:

$$\nabla \rightarrow \nabla_{\mathbf{x}} + \frac{1}{\varepsilon} \nabla_{\mathbf{y}}. \quad (1)$$

We assume that the microscale variable is defined in a unit cuboidal cell denoted Ω and that the solution is periodic within each periodic cell Ω . We denote the fluid and solid regions of the cell as Ω_s and Ω_f , respectively.

2 Governing Equations and non-Dimensionalisation

We will assume that inertia is negligible and so the governing equations will be the incompressible Stokes equations. We will denote dimensional variables with a $'$, as we will soon resort to dimensionless variables. The Stokes equations read:

$$\nabla' p' = \mu \nabla'^2 \mathbf{u}', \quad (2)$$

$$\nabla' \cdot \mathbf{u}' = 0, \quad (3)$$

where $\mathbf{u}$ denotes the instantaneous fluid velocity (no averaging), μ is the shear viscosity and p denotes the pore pressure.

We non-dimensionalise equations (2)-(3) by choosing the pressure gradient to balance the viscous dissipation, which is assumed to dominate on the pore scale (r_p):

$$\frac{P}{L} \sim \mu \frac{U}{r_p^2}, \quad (4)$$

where P and U are the pressure and velocity scales respectively. This gives a pressure scaling of:

$$P \sim \mu \frac{UL}{r_p^2}. \quad (5)$$

So if we scale p' with P and $\mathbf{u}'$ with U , we get the following non-dimensionalised equations:

$$\nabla p = \varepsilon^2 \nabla^2 \mathbf{u}, \quad (6)$$

$$\nabla \cdot \mathbf{u} = 0. \quad (7)$$

The no-slip boundary condition gives:

$$\mathbf{u} = \mathbf{0} \quad \text{on } \partial V_s. \quad (8)$$

3 Multiple Scales Analysis

We now use the method of multiple scales and transform the derivatives in equations (2) - (3) according to equation (1). This gives:

$$(\varepsilon \nabla_x + \nabla_y) p = \varepsilon (\varepsilon \nabla_x + \nabla_y)^2 \mathbf{u}, \quad (9)$$

$$(\nabla_y + \varepsilon \nabla_x) \cdot \mathbf{u} = 0. \quad (10)$$

Using a multiple scales approach, we shall pose the following asymptotic expansions for $\mathbf{u}$ and p , which are valid in the limit $\varepsilon \rightarrow 0$:

$$\mathbf{u} \sim \mathbf{u}_0(\mathbf{x}, \mathbf{y}) + \varepsilon \mathbf{u}_1(\mathbf{x}, \mathbf{y}) + \dots, \quad (11)$$

$$p \sim p_0(\mathbf{x}, \mathbf{y}) + \varepsilon p_1(\mathbf{x}, \mathbf{y}) + \dots \quad (12)$$

Inserting expansions (11)-(12) to equations (9)-(10), we get the following $\mathcal{O}(1)$ equations:

$$\nabla_y p_0 = 0, \quad (13)$$

$$\nabla_y \cdot \mathbf{u}_0 = 0. \quad (14)$$

with the no-slip condition:

$$\mathbf{u}_0 = \mathbf{0} \quad \text{on } \partial V_s \quad (15)$$

In particular, equation (13) implies that:

$$p_0 = p_0(\mathbf{x}). \quad (16)$$

At $\mathcal{O}(\varepsilon)$, we get the following set of equations:

$$\nabla_x p_0 + \nabla_y p_1 = \nabla_y^2 \mathbf{u}_0, \quad (17)$$

$$\nabla_y \cdot \mathbf{u}_1 + \nabla_x \cdot \mathbf{u}_0 = 0. \quad (18)$$

In particular, equation (17) can be viewed as a forced Stokes equation with the forcing being $\nabla_x p_0$. From the linearity of the Stokes equations, we can infer that $\mathbf{u}_0$ and $\nabla_y p_1$ must be linear in the forcing - $\nabla_x p_0$, i.e.

$$\mathbf{u}_0 = -\underline{\underline{\mathbf{K}}} \cdot \nabla_x p_0, \quad (19)$$

$$p_1 = \mathbf{A} \nabla_x p_0, \quad (20)$$

where $\mathbf{A}$ and $\underline{\underline{\mathbf{K}}}$ are tensors of rank 1 and 2, respectively. These tensors will be dependent on the microstructure, but their exact form will not be relevant to the subsequent homogenisation.

4 Homogenisation

We now proceed to deriving a set of effective equations that are valid on the macroscale by averaging on the microscale. We define the following cell average as:

$$\langle \cdot \rangle = \frac{1}{|\omega|} \int_{\omega} \cdot d\mathbf{y}, \quad (21)$$

where $|\omega|$ denotes the volume of the unit cell and $d\mathbf{y} = dy_1 dy_2 dy_3$ in 3 dimensions.

We start by taking the cell average of equation (19) and using equation (16) we have:

$$\langle \mathbf{u}_0 \rangle = -\langle \underline{\underline{\mathbf{K}}} \rangle \cdot \nabla_x p_0. \quad (22)$$

To take the cell average of equation (18), we make use of the divergence theorem as follows:

$$\langle \nabla_y \cdot \mathbf{u}_1 \rangle = \frac{1}{|\omega|} \int_{\omega} \mathbf{u}_1 d\mathbf{y} = \frac{1}{|\omega|} \int_{\partial\omega} \mathbf{u}_1 \cdot \mathbf{n} dS_y, \quad (23)$$

where $\partial\omega$ is the boundary of the unit cell and $\mathbf{n}$ denotes the outer unit normal to the cell. Since we have assumed that $\mathbf{u}_1$ is periodic on microscale, we have that $\mathbf{u}_1$ takes the same value on opposite faces of the cuboidal cell ω . However on each face, the sign of the unit normal $\mathbf{n}$ reverses and thus we have $\langle \nabla_y \cdot \mathbf{u}_1 \rangle = \mathbf{0}$. Then using the fact that ω is independent of $\mathbf{x}$, we get the following homogenised equation:

$$\nabla_x \cdot \langle \mathbf{u}_0 \rangle = 0. \quad (24)$$

In order to simplify equations (22)-(24), we make the assumption that the medium is homogeneous and isotropic, which mathematically corresponds to the fact that the tensor $\langle \underline{\underline{\mathbf{K}}} \rangle$ is isotropic, i.e.

$$\langle \underline{\underline{\mathbf{K}}} \rangle = K \underline{\underline{\mathbf{I}}}, \quad (25)$$

where K is a scalar and $\underline{\underline{\mathbf{I}}}$ denotes the 3×3 identity matrix. We also denote $\mathbf{U}_0 = \langle \mathbf{u}_0 \rangle$. Using this assumption and notation, we obtain the following equations once we go back to dimensional variables:

$$\mathbf{U} = -\frac{K}{\mu} \nabla_x p_0, \quad (26)$$

$$\nabla_x \cdot \mathbf{U} = 0. \quad (27)$$

A further simplification can be made by expressing $\mathbf{U}$, which is a leading order velocity averaged over the porescale with $\mathbf{u}_h$, which is a velocity averaged over the entire fluid-solid matrix, by:

$$\mathbf{U} = \frac{1}{\phi} \mathbf{u}_h, \quad (28)$$

where ϕ is the porosity, defined as:

$$\phi = \frac{|\omega_f|}{|\omega|}, \quad (29)$$

where $|\omega_f|$ denotes the volume of the fluid portion of the cell.

References

- [1] M. H. Holmes. *Introduction to Perturbation Methods*. International series of monographs on physics. Clarendon Press, 1981.